

Lab Assignment 9: BJT Characterization

Revision: October 30, 2018

Summary

The bipolar junction transistor (BJT), although supplanted by CMOS as the dominant transistor technology in integrated circuits, remains an important device for discrete-component printed circuit board designs, and also has important design advantages for very high frequency and/or high speed circuits. In this lab, we use curve tracers to plot the characteristic curves of an npn BJT. From the curves, we derive key BJT parameters, such as the DC current gain β (also known as h_{FE} on device data sheets), the small signal current gain β_{ac} (also known as h_{fe} on device data sheets), the reverse DC current gain β_r , the output resistance r_o , the Early voltage V_A , and the input resistance r_{π} .

Learning Outcomes: After completing this lab, you should be able to:

- Use curve tracers to plot i_C - V_{CE} and i_B - V_{BE} characteristic curves for an npn transistor.
- From the characteristic curves, find BJT parameters β (also known as h_{FE}), β_{ac} (also known as h_{fe}), β_r , r_o , V_A , and r_{π} .

Required Equipment

- EE352 analog parts kit
- Tek 575 curve tracer (or equivalent)
- Tek 571 curve tracer (or equivalent)

I. BJT i_C - v_{CE} Characteristic Curves

In this part of the lab, you will use the Tek 575 and 571 curve tracers to plot the i_C - v_{CE} characteristic curves for the 2N3904 npn transistor from your parts kit, and use the curves to find the following transistor parameters:

- the DC current gain β (also known as h_{FE} on device data sheets),
- the small signal current gain β_{ac} (also known as h_{fe} on device data sheets),
- the output resistance r_o (looking into the collector),
- the Early voltage V_A , and
- the reverse DC current gain β_r .

Figure 1 shows the pinouts of the 2N3904 npn transistor from your parts kit.

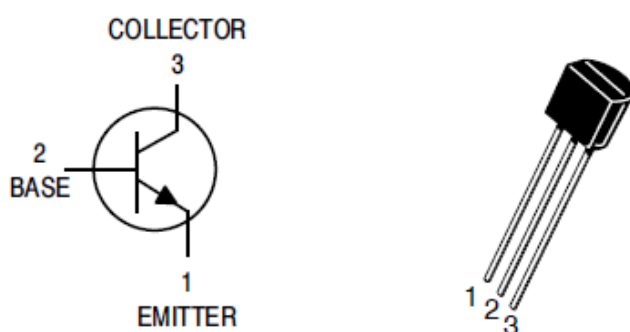


Figure 1. 2N3904 npn transistor. Maximum $V_{CE} = 40$ volts. Maximum $I_C = 200$ mA.

Lab Procedures:

- Before using the Tek 575 curve tracer, make sure it has warmed up for at least 15 minutes.
- Connect the 2N3904 transistor to the left hand transistor socket of the Tek 575; see Fig. A1 in the Appendix for a picture. Initially the toggle switch should be in the middle position (as in Fig. A1).
- In the selector sweep section (lower left) select the 0-20V, npn voltage settings. Initially set the peak collector voltage $V_C = 0$ V, and select a $50\ \Omega$ load resistor. See Appendix Fig. A2 for a picture.
- In the base step generator section (lower right) select repetitive, 4-12 steps, + polarity, 240 steps/sec and I_B steps of $10\ \mu\text{A}$. See Appendix Fig. A3 for a picture.
- In the vertical and horizontal scale section (upper right), display I_C on the vertical at 2mA/div , and V_{CE} on the horizontal at 2 volts/div . See Appendix Fig. A4 for a picture.
- Flip the transistor socket toggle switch to the left position. Slowly turn the V_C knob (peak collector voltage at lower left) all the way to the right, to sweep out a set of i_C - v_{CE} curves. Adjust the vertical and horizontal position knobs to put the origin in the lower left corner of the grid on the display; use the “zero check” on the horizontal and vertical scale section to locate the origin (see Fig. A4). Use the “zero current” switch in the base step generator

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section (see Fig. A3) to find the 0 base current line, which should lie along the lowest screen grid line. You should have enough curves to span a current range from 0 to at least 10 mA; to add more curves, turn the “4-12 steps” knob to the right (see Fig. A3). Use your cell phone to take a clean picture of these curves, which should resemble Fig. 2(b) below.

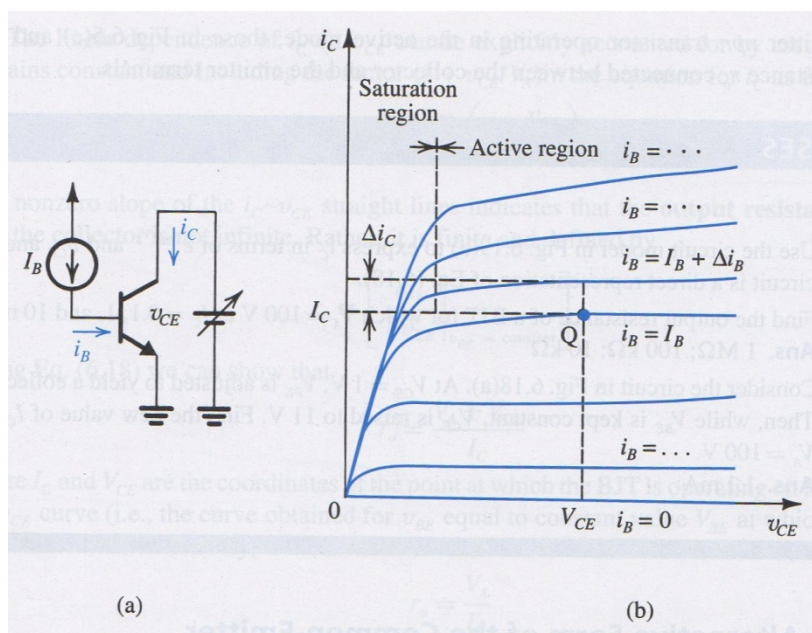


Figure 2 (from Sedra and Smith Fig. 6.20). (a) Schematic setup of i_C - v_{CE} curve generation. (b) Example set of i_C - v_{CE} curves, showing an operating point Q on one of the curves.

7. Turn the V_C sweep knob back to zero. Turn the “peak volts range” knob (just above the V_C knob – see Fig. A2) from 0-20 volts to 0-200 volts. Turn the horizontal scale to 5 volts/div (Fig. A4). *Very slowly* turn the V_C sweep knob to the right, to observe non-linear behavior of the i_C - v_{CE} curves at high values of v_{CE} . *Very important: do not exceed $V_{CE} = 40$ volts (8 horizontal divisions), or you may burn out your transistor.* Take a clean picture of this set of curves with your cell phone, and include it with your lab report. After taking your picture, make sure to return the V_C sweep knob back to zero, and turn the “peak volts range” knob back to 0-20 volts. *After completing step 7, release your Tek 575 to the next lab group waiting to use it.*
8. Load your cell phone image from step 5 into your laptop or tablet, and blow it up so that you can take measurements. Select four operating points Q, and for each operating point compute $\beta = I_C/i_B$, as in Fig. 2. You can find the base current i_B by counting the number of curves from the bottom, since each curve represents a base current step of $10\mu\text{A}$. Two of your operating points should be at $V_{CE} = 1\text{V}$, and the other two at $V_{CE} = 10\text{V}$. For each V_{CE} value, you should select two different curves (with different values of i_B) that intersect the given V_{CE} value at $I_C \approx 2\text{ mA}$ and $\approx 10\text{ mA}$. (Don’t worry if the intersections are not exactly at these I_C values; just get as close as you can). For each of the four operating points, record the voltages, currents (including the *actual* I_C values), and β values so as to fill out the table below, and record it in your lab notebook. Include the table in your lab report, and comment on the range of β values. In your lab report, compare the value of β you compute at $V_{CE} = 1\text{V}$

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and $I_C \approx 10$ mA to the minimum and maximum values of h_{FE} in the 2N3904 data sheet for that operating point; does your β value lie between the minimum and maximum?

V_{CE} (V)	I_C (mA)	I_B (μ A)	β
1	≈ 2		
1	≈ 10		
10	≈ 2		
10	≈ 10		

- Starting from your $(V_{CE}, I_C) = (10V, 2 \text{ mA})$ operating point, select the next highest curve and use it to compute the small signal current gain $\beta_{ac} = \Delta i_C / \Delta i_B$, as shown in Fig. 2(b) above, and record it in your lab notebook. Compare your computed value of β_{ac} with the value of h_{fe} in the 2N3904 data sheet, and with the value of β you computed at the $(V_{CE}, I_C) = (10V, 2 \text{ mA})$ operating point.
- Using your picture from step 6, measure the slope of the curve at your $(V_{CE}, I_C) = (10V, 2 \text{ mA})$ operating point. Compute the output resistance $r_o = 1/\text{slope}$ and record it in your lab notebook. Make sure that you compute r_o in units of Ohms = Volts/Amps.
- From your value of r_o and its operating point, compute the Early voltage $V_A = r_o \times I_C - V_{CE}$, and record it in your lab notebook.
- Save and mark the transistor you used for steps 1-11; it will be used for the remaining steps in this Lab 9 assignment, and it will be used in Lab 10.
- Tek 571 curves:** Using the Tek 571 curve tracer, make a set of i_C - V_{CE} curves for comparison with your Tek 575 curves. A picture of the transistor connection is shown in Fig. A5. The Tek 571 settings to use are: NPN, $V_{CEmax} = 20 \text{ V}$, $I_{Cmax} = 20 \text{ mA}$, $I_B/\text{step} = 10\mu\text{A}$, Steps = 6, $R_{load} = 0.25 \Omega$, and $P_{max} = 0.5 \text{ Watt}$. Take a clean picture of the Tek 571 curves and include it in your lab report.
- Reverse β measurement:** Back at the Tek 575, center the transistor toggle switch, and plug your transistor into the left hand socket with the C and E connections reversed, so that the emitter pin of the transistor goes into the collector socket, and the collector pin goes into the emitter socket. The procedure is identical to steps 2-6 above, except that you should set the collector voltage horizontal scale to 0.5 Volts/div, and the collector current vertical scale to 0.05 mA/div (see Fig. A4). Slowly increase the collector sweep voltage, and do not go higher than about 5 volts. Take a clean picture of the resulting family of curves. *After taking your picture, release the Tek 575 to the next lab group waiting for it.* Using your blown up picture, select an operating point with $V_{CE} = 2.5$ volts and I_C of your choice; measure and record your operating point (V_{CE}, I_C) and then compute and record the reverse beta $\beta_r = I_C / I_B$.

II. BJT i_B - V_{BE} Characteristic Curves

In this part of the lab, you will use the Tek 575 curve tracer to plot the i_B - V_{BE} characteristic curve for the 2N3904 npn transistor, and use the curve to find the input resistance r_{π} .

Lab Procedures:

- Switch the display section (upper right) to base current I_B (vertical) vs. base to emitter voltage V_{BE} (horizontal), both on blue scales. The vertical scale corresponds to the I_B steps

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selected (lower right); the I_B steps should be set at $10\mu\text{A}$ per step. Set the horizontal scale to 0.1 volts/div . See Figure A6 for a picture.

2. Create the curve by turning the “4-12” steps knob to the right; there should be one point on the curve per step. Important: use the zero check amplifier calibration switches below the horizontal and vertical scale selector knobs (see Fig. A4) to ensure that origin of your curve is at lower left of the grid on the screen. Use the “step zero” knob on the lower right of the base step generator (see Fig. A3) to extend the starting point of your curve to the origin.
3. Once you have your curve correctly oriented on the screen with the origin at the lower left corner of the grid, take a clean picture of the curve, *and then release the Tek 571 to the next lab group waiting for it*. After loading your curve to your laptop or tablet, measure the slope at an operating point of $I_B = 20\mu\text{A}$. Compute the input resistance $r_\pi = 1/\text{slope}$. Make sure that the units of r_π are Ohms.
4. *Show your Tek 575 curve pictures and your calculations of β , β_{ac} , r_o , V_A , β_r , and r_π to the TA and have him or her initial your lab checklist.*

Post Lab Assignment:

Using LTSPICE or Orcad, obtain the I-V characteristics curve using primary and secondary DC simulation to obtain a graph similar to the one shown in Figure 2; it is sufficient to produce a curve for a single value of i_B . Use Q2N3904 from the components or eval library.

Lab Report:

In your lab report, provide a summary of the results of this lab assignment. You should include, at a minimum, all items indicated on the lab checklist. Append the lab checklist sheet with Laboratory Instructor's initials indicating completed lab demos to your report.

Appendix A: Curve Tracer Setup Pictures



Figure A1. Two 2N3904 transistors in the left and right sockets of the Tek 575. With toggle switch in center position as shown, neither device is connected to the instrument's measurement circuits. Note position of central rotator switch at "Emitter Grounded".



Figure A2 Collector sweep settings for the Tek 575. Peak volts knob at lower left should initially be set to 0.



Figure A3. Base step settings for the Tek 575.

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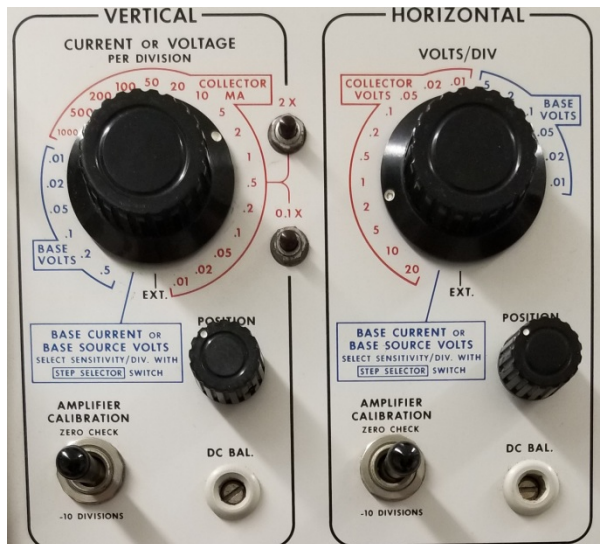


Figure A4. Vertical and horizontal scale settings for I_C - V_{CE} curves.



Figure A5. Tek 571 transistor connection.



Figure A6. Tek 575 scale and base step settings for I_B - V_{BE} curve.

Lab 9 Checklist (50 total points)

Name: _____

Lab title and introduction

(3 pts)

- Lab title, your name, date, and lab partner.
- Brief introduction (two or three sentences) explaining the purpose of this lab.

BJT Characteristics (47 pts total)

1. Table including typical range of values and your calculated values for the 2N3904 based on measurements of the 575 curve tracer graphs. Calculated values should be at the operating points specified in the lab assignment.

Note that you must include an explanation (either in words, equations, or both) of how you obtained your calculated values to receive full credit for this part.

- | | |
|--|---------|
| (a) DC current gain β (for all four operating points). | (5 pts) |
| (b) Small signal current gain β_{ac} . | (5 pts) |
| (c) Collector resistance r_o . | (5 pts) |
| (d) Early voltage V_A . | (5 pts) |
| (e) Reverse β . | (2 pts) |
| (f) Input resistance r_{π} . | (5 pts) |

_____ **DEMO:** Have a TA initial this sheet, indicating that he/she has observed your TEK 575 curves and the calculations of parts 1(a)-(f). (5 pts)

2. Picture of characteristic curves from the 571 curve tracer; compare to Tek 575 curves. (5 pts)
3. Compare your expected and experimental results; give reasons for any discrepancies. (5 pts)
4. Post lab assignment. (5 pts)